Md Ahsanul Haque, Ph.D.
MAT-120

Homework-1

Date: July 9, 2026

Name

Section

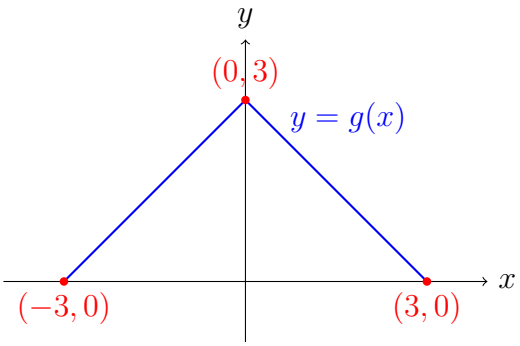
Instructions:

- Write your full name and section number inside the provided box.
- This assignment contains exactly 5 problems.
- Use correct mathematical notation and clearly show all logical steps.

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1. (10 points)
   (a) Given the function $f(x) = -\sqrt{9 - x^2}$, determine:
      i. The domain and range of $f(x)$.
      ii. Whether $f(x)$ is one-to-one (justify your answer algebraically or graphically).
      iii. The inverse function $f^{-1}(x)$ if it exists, including its restricted domain and range.

   (b) The graph of a piecewise linear function $y = g(x)$ is shown below. Sketch the graphs of the following transformations, clearly labeling the new coordinates of the three key points:



      i. $y = g(x + 1) - 2$
      ii. $y = -\frac{1}{2}g(x)$
      iii. $y = g(-2x)$

2. (10 points)
   (a) Solve for x exactly. Be sure to verify your solutions against the domain of the original equation:

$$\log_3(x+4) - \log_3(x-2) = 1 + \log_3(2)$$

   (b) Given the functions $f(x) = 5\ln(2x-1)$ and $g(x) = e^{x/5}$:
      i. Find the composite function $(f \circ g)(x)$ and state its domain.
      ii. Find the composite function $(g \circ f)(x)$ and state its domain.

3. (10 points) A radioactive isotope decays exponentially according to the model

$$M(t) = M_0 e^{-rt}$$

   where t is measured in days. The isotope has a half-life of 15 days, and the initial sample mass is 800 grams.
   (a) Determine the exact value of the decay constant r.
   (b) How many days will it take for the sample to decay to a mass of exactly 50 grams?
   (c) Find the average rate of decay of the substance between $t = 15$ days and $t = 30$ days. Include units in your answer.

4. (10 points)
   (a) Consider the rational function $f(x) = \dfrac{5x-2}{3x+4}$. Find its inverse function $f^{-1}(x)$ and algebraically verify your result by showing that $f(f^{-1}(x)) = x$.
   (b) Analyze and formally state the symmetry (even, odd, or neither) of the following functions. Show all algebraic testing steps.
      i. $h(x) = x^2 \sin(x)$
      ii. $k(x) = \ln(x^2+1)$

5. (10 points)
   (a) A manufacturing plant models its total weekly cost (in dollars) using the function $C(x) = 500 + 10x + 0.1x^2$, where x represents the number of units produced. The average cost function is defined as $\bar{C}(x) = \frac{C(x)}{x}$.
      i. Find the practical mathematical domain of $\bar{C}(x)$ in the context of manufacturing.
      ii. Write the explicit formula for $\bar{C}(x)$ and evaluate $\bar{C}(10)$. Briefly interpret what this numerical value means for the plant.

   (b) Let $f(x)$ and $g(x)$ both be odd functions that are defined for all real numbers. Let $H(x) = f(x)g(x)$ be their product. Prove algebraically that $H(x)$ must be an even function.